

A_7 curves and accessory degrees

A consolidated mathematical reference

1 October 2026 · recovered checkpoints, proof mechanisms, and structural continuation

Scope. Curves are smooth, connected and projective over $\mathbb{C}$ unless stated otherwise. Finite actions on connected curves are faithful. A pencil is a map to $\mathbb{P}^1$ up to target automorphism; its underlying class and its two-dimensional section space must not be conflated. Picard notation is additive. Classical inputs are explicitly identified; proofs below are research proofs, not claims of independent refereeing or bibliographic priority.

The new conclusion. For the field-theoretic accessory convention used in this project, allowing the Galois group to decrease after the accessory,

$$\boxed{\text{ed}_{\mathbb{C}}(A_7; \leq 59) > 1, \quad \text{ed}_{\mathbb{C}}(A_7; \leq 60) = 1.}$$

Sections 1–3 give the full argument. The key object is a *linearized moving line bundle*, not merely a pencil or an invariant Picard class. The upper construction uses the index-15 $\text{PSL}_2(7)$ subgroup and its degree-four canonical Klein quartic. It does not assert a degree-60 compression that retains connected full- A_7 monodromy.

Result or mechanism	Status and precise scope
Accessory threshold 60	New proof here, from classical algebraic geometry and finite-group data.
Ramification transport	Algebraic proof: $\text{gon}(C/\tau) \geq 10$ for signature $(2, 4, 7)$.
Weighted pencil geometry	Sharp independent-triple bound; $\text{gon}(C) \geq 25$ for $g(C) \geq 266$.
Spectral bounds 24/12	Prior audited computational result for $(2, 4, 7)$; numerical recovery is not fully self-contained in the recovered PR.
Conformal optimization	Strict invariant conformal improvement; existence of an equivariant, possibly conical maximizer. No exact value or fixed 14_a target is proved.
Arithmetic	Exact Belyi and resolvent function fields; explicit elliptic coordinates, ranks and torsion remain uncomputed.
Unrestricted towers	No lower bound for resolvent degree is established by these results.

Provenance. This reference consolidates the four “DAY 1” PDFs and the repository [PR #2](#), head [284e1e24eb31477220cd3919840b4bb5758025a7](#). Earlier weaker bounds are replaced by their strongest retained versions. Distinct mechanisms, including superseded arithmetic reductions, are retained in compact form. The accessory theorem is developed in this continuation.

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1 Linearized moving degree and compression

Let V be a faithful linear representation of a finite group G , $L = \mathbb{C}(V)$ and $K = L^G$. An *accessory* is an arbitrary finite extension F/K . A curve compression is descent of the resulting G -torsor to a subfield of transcendence degree at most one. If LF/F has connected monodromy $H \leq G$, this means a faithful H -stable curve field $E \subset LF$ with $LF = FE$. The induced G -curve may be disconnected; its component actions must be generically free. Merely faithful action on the set of components is insufficient.

Definition 1.1. Write $\mu(G)$ for the least degree of a G -linearized base-point-free line bundle A on a faithful connected G -curve with $h^0(A) \geq 2$. Set $\mu(1) = 1$. Write $a(G)$ for the least accessory-degree bound allowing curve compression of a generic G -torsor. These symbols are working notation for this reference.

Theorem 1.2 (Linearized correspondence). *Suppose $W \dashrightarrow V$ is an irreducible G -equivariant generically finite cover of degree d , and $W \dashrightarrow C$ is a dominant G -map to a faithful curve. Then C has a moving G -linearized line bundle of degree e for some $e \mid d$. In particular $\mu(G) \leq d$ after removing its fixed divisor.*

Proof. Let $Z \subset V \times C$ be the closure of the joint image. It is an integral G -stable codimension-one subvariety, generically finite of degree $e \mid d$ over V . The product is smooth, so Z is Cartier and $\mathcal{O}(Z)$ has its canonical G -linearization. Restrict this *bundle* to the fixed fibre $\{0\} \times C$, obtaining a linearized bundle A .

Homotopy invariance $\text{Pic}(V \times C) \simeq \text{Pic}(C)$ shows that all fibre restrictions have the same underlying class. On a general fibre this is the class of an effective divisor of degree e , so $\deg A = e$. Those divisors move: if their class had only one effective representative, the image of Z in C would be finite, contradicting dominance. Hence $h^0(A) \geq 2$. The fixed divisor of the complete series is G -invariant; subtracting it preserves linearization and produces a base-point-free moving bundle. $\square$

Remark 1.3. The whole fibre $\{0\} \times C$ may lie in Z . This does not invalidate the proof: the canonical section may restrict to zero, but the line bundle still restricts and retains its linearization and degree. The joint-image degree e accounts for a possible nonbirational map $W \dashrightarrow Z$.

Proposition 1.4 (Incidence converse). *A base-point-free G -linearized bundle of degree m with at least two sections gives an irreducible equivariant degree- m cover of some faithful linear G -variety, dominating C .*

Proof. For $S = H^0(C, A)$, the incidence variety

$$I_A = \{(s, p) \in S \times C : s(p) = 0\}$$

is the total space of the kernel of the surjective evaluation map $S \otimes \mathcal{O}_C \rightarrow A$. Thus it is irreducible and dominates C ; over a general $s \in S$ its fibre has length m . Both projections are equivariant. If S is not faithful, replace the base by $S \oplus V_0$ for a faithful representation V_0 , and the incidence variety by $I_A \times V_0$. The degree remains m . $\square$

Thus $\mu(G)$ is exactly the minimum degree for a connected full- G curve compression when the faithful linear model is allowed to vary. A stronger demand that the complete linear series be birational or projectively faithful is *not* part of its definition.

Theorem 1.5 (Subgroup induction formula). *Under the accessory convention above,*

$$\boxed{a(G) = \min_{H \leq G} [G : H] \mu(H).} \tag{1.1}$$

Proof. For the lower bound, let F/K be an accessory giving a curve compression and write $L \cap F = L^H$. Galois theory gives

$$[F : K] = [G : H]e, \quad [LF : L] = e = [F : L^H].$$

The normalization of V in LF is an irreducible H -equivariant degree- e cover. The compressed curve field is faithful for H , since its compositum with F is LF . Theorem 1.2 therefore gives $\mu(H) \leq e$. If $H = 1$, this last inequality is automatic and covers compression to a point as well.

For the upper bound choose an H -curve with a linearized bundle A of degree $m = \mu(H)$, and put $S = H^0(C, A)$. Take the faithful G -representation $U = \text{Ind}_H^G S \oplus V_0$, with V_0 faithful. Restricted to H , it has an equivariant projection to its identity-coset summand S . Pull the incidence variety I_A back along this projection. The resulting irreducible H -variety J is a degree- m cover of U with a dominant H -map to C .

Put $L' = \mathbb{C}(U)$, $K' = (L')^G$, and $F = \mathbb{C}(J)^H$. Since H acts faithfully on L' , its fixed-field theorem gives $L'F = \mathbb{C}(J)$. Degree counting gives

$$[F : K'] = \frac{[\mathbb{C}(J) : L'][L' : K']}{[\mathbb{C}(J) : F]} = \frac{m|G|}{|H|} = [G : H]m.$$

This accessory compresses the component H -torsor to C ; induction yields the required G -torsor compression. Essential dimension relative to bounded-degree covers is independent of the faithful linear model [1], so this suffices for $a(G)$. Equivalently, twist C, A by an H -torsor and adjoin a generic section zero. $\square$

2 Two degree lattices and a representation bound

Theorem 2.1 (Equivariant Picard degree lattice). *Let $e_1, \dots, e_r$ be the inertia orders of $C \rightarrow C/G$, and put*

$$N_G(C) = \frac{|G|}{\text{lcm}(e_1, \dots, e_r)}, \quad \text{lcm}(\emptyset) = 1.$$

Then the degree image of the group of G -linearized line bundles is exactly

$$\deg \text{Pic}_{\text{lin}}^G(C) = N_G(C)\mathbb{Z}. \quad (2.1)$$

If $s = \exp H^2(G, \mathbb{C}^)$, every invariant class $A \in \text{Pic}(C)^G$ satisfies*

$$N_G(C) \mid s \deg A, \quad \deg A \in \frac{N_G(C)}{\gcd(N_G(C), s)}\mathbb{Z}. \quad (2.2)$$

Proof. A linearized bundle has an invariant rational section by Hilbert 90 for $\mathbb{C}(C)/\mathbb{C}(C)^G$. Its divisor is invariant. Conversely, invariant divisors have canonical linearizations. Their degrees are generated by the orbit sizes $|G|, |G|/e_1, \dots, |G|/e_r$, with $\gcd N_G(C)$. For an invariant class, choices of pullback isomorphisms have a scalar two-cocycle obstruction in $H^2(G, \mathbb{C}^*)$. Its s th tensor power is linearizable, proving (2.2). $\square$

The second formula is a necessary lattice, not generally an equality. When N/s is not integral, the expression $(N/s)\mathbb{Z}$ must be intersected with $\mathbb{Z}$; (2.2) is the unambiguous integral formulation. These are standard equivariant Picard and descent arguments; their use in the correspondence theorem is the new mechanism here.

Action	$ G $	$N_G(C)$	s	necessary invariant-class divisor
$A_7, (2, 4, 7)$	2520	90	6	15
$\text{PSL}_2(13), (2, 3, 7)$	1092	26	2	13

For a free cyclic group of order n , every invariant class has degree divisible by n : rescale a lift of its generator until its n th power is the identity, then descend. This can be stronger than the coarse Schur-exponent lattice for an ambient group.

Proposition 2.2 (Linearized Castelnuovo obstruction). *Let G be nonabelian simple, let $q(G)$ be the smallest dimension of a nontrivial complex linear representation, and suppose A is a moving linearized bundle of degree $m < |G|$. There is a faithful G -curve Y with a birational nondegenerate model of degree $n \leq m$ in $\mathbb{P}^r$, where $r \geq q(G) - 1$, and*

$$N_G(Y) \mid n, \quad g(Y) \leq \pi(n, q(G) - 1). \quad (2.3)$$

Here the ordinary Castelnuovo number is

$$\pi(n, r) = (r - 1) \binom{k}{2} + kt, \quad n - 1 = (r - 1)k + t, \quad 0 \leq t < r - 1.$$

Proof. Remove the fixed divisor, map by the complete series, and normalize its image to Y . The kernel of G on Y has order dividing $\deg(C \rightarrow Y) \leq m$. Simplicity and $m < |G|$ make that kernel trivial. Thus the genuine representation on sections is nontrivial and has dimension at least $q(G)$. The image is nondegenerate; its degree n is at most m . The pulled-back hyperplane bundle on Y is linearized. Apply (2.1) and Castelnuovo, whose bound decreases with the ambient dimension. $\square$

An arbitrary invariant line bundle supplies only a projective representation. That distinction prevents replacing $q(G)$ here by the dimension of a convenient projective representation, or using a gonial pencil as if it were linearized.

3 The exact A_7 accessory threshold

The finite-group inputs are $|A_7| = 2520$, exponent 420, Schur multiplier exponent 6, and smallest nontrivial linear degree 6; for A_6 the smallest degree is 5. The maximal subgroups of A_7 are A_6 , two classes of $\mathrm{PSL}_2(7)$, S_5 , and $(A_4 \times C_3) : 2$ [2, 3]. The ordinary irreducible dimensions, which also certify the linear-degree inputs, are

$$\begin{array}{l|l} A_7 & 1, 6, 10, 10, 14, 14, 15, 21, 35 \\ A_6 & 1, 5, 5, 8, 8, 9, 10. \end{array}$$

3.1 The small-genus table

Riemann–Hurwitz gives, with $h = g(C/A_7)$,

$$g(C) - 1 = 1260 \left(2h - 2 + \sum_i (1 - 1/e_i) \right). \quad (3.1)$$

The minimum faithful genus is 136. Genus zero and one are excluded by their finite automorphism groups. One can check the remaining claim without a classification of curves. Below 136, the only hyperbolic triangle candidates are $(2, 3, 7)$, $(2, 4, 5)$, $(2, 4, 6)$, $(3, 3, 4)$, $(2, 5, 5)$. The natural seven-sheeted quotient requires total permutation ramification at least 12; this excludes the three containing order two but not $(2, 3, 7)$. Direct enumeration of seven-letter permutations excludes the other two: fixed first-class representatives and all second generators give the following subgroup orders.

Orders	generated subgroup order	number of pairs tested with that order
$(2, 3, 7)$	168	96
$(3, 3, 4)$	168; 360	72; 90

No subgroup has order 2520. A generating $(2, 4, 7)$ triple realizes genus 136. The count includes both order-three conjugacy classes. In particular, the claim that *all* $(3, 3, 4)$ pairs generate only order 168 is unnecessary and inaccurate.

For $136 \leq g \leq 325$, a quotient of genus one with any branching has $g \geq 631$; an unbranched genus-one action would have abelian deck group. Quotient genus at least two gives $g \geq 2521$. On a rational quotient, five branch points give $g \geq 631$. The only four-point candidates are $(2, 2, 2, 3)$ and $(2, 2, 2, 4)$, both failing the seven-sheeted test. The same test excludes $(2, 5, 6)$ and $(2, 6, 6)$. Enumerating the remaining triples in (3.1) leaves:

Signature	g	N	Signature	g	N
$(2, 4, 7)$	136	90	$(3, 3, 7)$	241	120
$(3, 3, 5)$	169	168	$(2, 7, 7)$	271	180
$(2, 5, 7)$	199	36	$(3, 4, 5)$	274	42
$(3, 3, 6)$	211	420	$(3, 4, 6)$	316	210
$(3, 4, 4)$	211	210	$(4, 4, 4)$	316	630
$(2, 6, 7)$	241	60			

Only necessary signature data are needed for the argument; existence in every row is not required. The enumeration and Castelnuovo arithmetic are reproduced by the companion exact checker.

Theorem 3.1. $\mu(A_7) \geq 60$.

Proof. Suppose $m \leq 59$ and apply Proposition 2.2. Every possible inertia order divides 420, so $6 \mid N_{A_7}(Y)$ and therefore $6 \mid n$. Thus $n \leq 54$ and

$$136 \leq g(Y) \leq \pi(54, 5) = 325.$$

In the table the only lattice possibilities with $1 \leq n \leq 54$ are $n = 36$ for signature $(2, 5, 7)$ and $n = 42$ for $(3, 4, 5)$. But respectively

$$199 > \pi(36, 5) = 136, \quad 274 > \pi(42, 5) = 190.$$

Both contradict Castelnuovo. □

3.2 Every subgroup wall below sixty

The proper subgroups of index below 60 have the types in the next table. This follows from the displayed maximal subgroups and the standard maximal-subgroup lists of A_6 , $\mathrm{PSL}_2(7)$ and S_5 : the only additional type of order greater than 42 is A_5 . Different conjugacy classes do not change the estimates.

H	$[A_7 : H]$	lower bound for $\mu(H)$	product lower bound
A_6	7	9	63
$\mathrm{PSL}_2(7)$	15	4	60
S_5	21	3	63
$(A_4 \times C_3) : 2$	35	2	70
A_5	42	2	84

Here are proofs of all the bounds needed, rather than assumptions about subgroup essential dimensions. For A_6 , a moving linearized bundle of degree at most eight would give a faithful curve of genus at most $\pi(8, 4) = 5$. Hurwitz gives genus at least six; genus zero and one are excluded by their finite automorphism groups. For $\mathrm{PSL}_2(7)$, a degree-at-most-three model in projective dimension at least two has genus at most one, again impossible.

For S_5 , degree at most two makes the curve rational, elliptic, or hyperelliptic. Elliptic automorphism groups are solvable. In the hyperelliptic case the central hyperelliptic involution meets the centerless S_5 trivially, so S_5 would inject into $\mathrm{PGL}_2(\mathbb{C})$. This is impossible, as is a rational action. The order-72 group cannot act faithfully on $\mathbb{P}^1$ because it contains C_3^2 ; thus degree one is impossible. Finally, a linearization of $\mathcal{O}_{\mathbb{P}^1}(1)$ for A_5 would give a nontrivial two-dimensional linear representation, which does not exist. The invariant conic realizes $\mu(A_5) = 2$.

Theorem 3.2 (Exact accessory threshold). *With accessories allowed to decrease monodromy,*

$$a(A_7) = 60, \quad \mathrm{ed}_{\mathbb{C}}(A_7; \leq 59) > 1, \quad \mathrm{ed}_{\mathbb{C}}(A_7; \leq 60) = 1.$$

Proof. Theorem 1.5, Theorem 3.1, and the subgroup table give $a(A_7) \geq 60$. For the reverse inequality take the index-15 subgroup $H = \mathrm{PSL}_2(7)$. Its Klein quartic has its canonical, genuinely H -linearized degree-four bundle. Hence $\mu(H) = 4$ and $a(A_7) \leq 15 \cdot 4 = 60$. Degree $60 < 2520$ cannot trivialize the generic A_7 -torsor, so the resulting essential dimension is one, not zero. $\square$

At degree 30, the index-15 intersection leaves a degree-two H accessory, which is excluded by $\mu(H) = 4$. Thus that wall changes the monodromy but is not the first successful degree. The degree-29 claim from the interrupted turn follows as a special case, with no spectral or three-pencil input.

4 Ramification transport

4.1 Fixed points and determinant saturation

For any finite action on a complex curve, point stabilizers are cyclic, since their derivative characters are faithful. In a branch orbit with cyclic inertia I , an element v has

$$|C_G(v)| |I \cap v^G| / |I|$$

fixed points. For an A_7 curve of signature $(2, 4, 7)$, this gives

$$g(C) = 136, \quad \deg R_v = 18 \quad (v^2 = 1), \quad g(C/\langle v \rangle) = 64.$$

An order-four element fixes two points, an order-seven element three; elements of orders three, five and six act freely. Involutions are the 105 double transpositions, with centralizer order 24. Its quotient by the central involution is $C_2 \times S_3$.

Lemma 4.1 (Saturation). *Let $f : Z \rightarrow \mathbb{P}^1$ have degree d , with pencil bundle A , and let v be an involution with r fixed points. If $r > 2d$, then $f \circ v = f$. If $r = 2d$ and $f \circ v \neq f$, then $[R_v] = A + v^*A$.*

Proof. For generating sections s_0, s_1 , the section $s_0 v^* s_1 - s_1 v^* s_0$ of $A \otimes v^* A$ vanishes at every fixed point. It is identically zero precisely when the two maps are equal. Otherwise its zero divisor has degree $2d$, and equality leaves no residual zeros. $\square$

If τ, μ are distinct commuting involutions and $\nu = \tau\mu$, their fixed sets are disjoint. On $D = C/\tau$ the induced involution satisfies

$$\pi_{\tau}^* R_{\bar{\mu}} = R_{\mu} + R_{\nu}. \tag{4.1}$$

It therefore fixes eighteen points. Three such involutions generate $C_G(\tau)/\tau \simeq C_2 \times S_3$. A map $D \rightarrow \mathbb{P}^1$ of degree at most eight would be invariant under all three, forcing its degree to be divisible by twelve. Thus $\mathrm{gon}(D) \geq 9$.

Theorem 4.2 (Local transport for A_7). *For every $(2, 4, 7)$ A_7 -curve and every involution τ , $\text{gon}(C/\tau) \geq 10$.*

Proof. A nine-pencil on D pulls back to a τ -invariant degree-18 pencil of class A . For each commuting $\mu \neq \tau$, its determinant against $f \circ \mu$ is nonzero: otherwise the Klein four group would fix f , forcing $4 \mid 18$. Saturation on (4.1) gives $A + \mu^*A = R_\mu + R_{\tau\mu}$. This also holds for each centralizer translate. Put

$$M = \sum_{h \in C_G(\tau)} h^*A, \quad T = 24R_\tau + 2M.$$

For every Klein group V through τ , summation gives $T = 24R_V$, where $R_V = \sum_{1 \neq v \in V} R_v$. In particular use

$$V_0 = \langle (12)(34), (13)(24) \rangle, \quad V_1 = \langle (12)(34), (12)(56) \rangle.$$

Their normalizers fix the same class T and generate A_7 : the first contains A_4 on 1234 and (567); the second contains (15)(26). Conjugation supplies A_4 on 3456, and the overlapping alternating groups generate A_6 , then A_7 . Thus T is invariant of degree $24 \cdot 54 = 1296$, contradicting free C_5 descent. Equivalently, the necessary invariant-class lattice is $15\mathbb{Z}$, and $15 \nmid 1296$. $\square$

In fact $\text{Aut}(C) = A_7$. Hurwitz bounds its index over A_7 by four. The coset action and simplicity force normality; the quotient action on C/A_7 fixes the three distinct branch orders, hence is trivial. Thus no degree-at-most-18 map from C has a degree-two first factor, even one not initially specified by A_7 .

4.2 Incidence, normalizer amalgamation, and defects

Use the left action $g \cdot A = (g^{-1})^*A$. Let Ω be an involution class with fixed-point count $r \equiv 2 \pmod{4}$, and join commuting distinct τ, μ when $\tau\mu \in \Omega$. Put $c = |C_G(\tau)|$.

Proposition 4.3 (Global transport). *If this graph is connected and C/τ has a degree- $r/2$ map, then $\text{Pic}^{3cr}(C)^G \neq \emptyset$. If it has b components, then $\text{Pic}^{3crb}(C)^G \neq \emptyset$.*

Proof. Choose conjugate pulled-back pencils A_τ of degree r and define $M_\tau = \sum_{h \in C_G(\tau)} h \cdot A_\tau$ and $T_\tau = cR_\tau + 2M_\tau$. The centralizer average removes the choice of conjugator. On an edge, saturation gives $T_\tau = c(R_\tau + R_\mu + R_{\tau\mu}) = T_\mu$. The family is equivariant. It is therefore constant on connected components; summing the component classes gives the second assertion. $\square$

One can sum over the distinct centralizer orbit of classes instead: if its size is s , the degree becomes $3sr$. The A_7 graph is connected, has degree eight, 420 edges, and 140 Klein triangles. Its four-letter supports connect through overlaps of size two. Connectivity is essential: for an A_5 action of signature $(2, 2, 2, 3)$ in genus six, the graph has five components, involution quotients have genus two and degree-three maps, although $5 \nmid 3 \cdot 4 \cdot 6$.

The local unequal-count form is equally useful. If a τ -invariant map has degree $m \equiv 2 \pmod{4}$, and Klein groups $V_i = \{1, \tau, \mu_i, \nu_i\}$ satisfy $\deg R_{\mu_i} + \deg R_{\nu_i} = 2m$, then

$$c[R_{V_i}] \text{ is independent of } i, \quad \deg(cR_{V_i}) = c(\deg R_\tau + 2m),$$

and the common class is fixed by $\langle N_G(V_i) \rangle$. A freely acting cyclic subgroup detects any incompatible degree. This is the reason local averaging is useful: averaging over all of G can multiply away the detecting prime.

Above saturation, the exact replacement is

$$A_\tau + \mu \cdot A_\tau = R_\mu + R_{\tau\mu} + E_{\tau,\mu}, \quad E_{\tau,\mu} \geq 0, \quad \deg E_{\tau,\mu} = 2(m - r).$$

After centralizer summation, with $\mathcal{E}_{\tau,\mu}$ the residual sum,

$$T_\tau - T_\mu = [\mathcal{E}_{\tau,\mu}] - [\mathcal{E}_{\mu,\tau}] \in \text{Jac}(C). \quad (4.2)$$

Closed paths have zero signed sum. These are necessary Abel-image incidence constraints; they do not prove a residual-balancing theorem for degree ten or eleven. For the two A_7 Klein classes, $\delta = [R_{V_0} - R_{V_1}]$ has infinite order or order divisible by five: $n\delta = 0$ would make nR_{V_0} invariant of degree $54n$.

Corollary 4.4 (Hurwitz example). *The genus-six involution quotients of the three genus-fourteen $\text{PSL}_2(13)$ Hurwitz curves have gonality exactly four.*

Proof. Here $r = 6$, $c = 12$, and two of the three A_4 normalizers through an involution generate the group. Saturation at degree three would force an invariant class of degree 216, contrary to free C_{13} descent. Good reflections generate S_3 and exclude degrees one and two. Their quotient curves have genus two; pulling back a hyperelliptic pencil gives degree four. The three resulting pencils are distinct, since sharing two reflection symmetries would force divisibility by six. $\square$

5 Pencil orbits and multigraded genus

5.1 Orbit fields and pair bounds

The many-functions Castelnuovo lemma says that equal degree- m functions generating $\mathbb{C}(C)$ force $g(C) \leq (m-1)^2$. Induct on the number of functions and use Castelnuovo–Severi:

$$g(X) \leq d_1 g(Y_1) + d_2 g(Y_2) + (d_1 - 1)(d_2 - 1)$$

when the product of the two maps is birational. The induction reduces to $m^2/e + em - 3m + 1 \leq (m-1)^2$ for $1 \leq e \leq m$.

For an A_7 pencil of degree $m < 60$, its stabilizer K acts on the target $\mathbb{P}^1$ with pointwise kernel K_0 , and $|K_0| \mid m$. The maximal-subgroup list and finite subgroups of PGL_2 give $|K| \leq 72$; order 72 requires $3 \mid m$. Thus the orbit has at least 35 members, or 42 if $3 \nmid m$. This is a statement about pencil subfields, not automatically about distinct Picard classes when a complete series has more than two sections.

For $m \leq 24$, the orbit functions generate the full field. Otherwise their orbit field has index $e \geq 2$, is faithful by simplicity, and is generated by degree $m/e \leq 12$ functions, contradicting the minimum genus 136. For a nonbirational pair, extend its compositum to a maximal proper compositum of orbit fields. If its index is e , adding one outside function and applying Castelnuovo–Severi gives

$$g(C) \leq B_e(m) := \frac{m^2}{e} + em - 3m + 1, \quad e \mid m, \quad 2 \leq e \leq m/2. \quad (5.1)$$

This avoids the invalid assumption that any arbitrary third makes a pair birational. When the initial pair itself has maximal index e , the two additional structures also give the useful comparison $g(C) \leq 2e(m/e - 1)^2 + (e - 1)^2$ whenever the two corresponding degree- e maps form a birational pair. Its hypotheses must be checked.

5.2 Weighted base points

Fix a birational pair of degree- m pencils. Its image $B \subset Q = \mathbb{P}^1 \times \mathbb{P}^1$ has type (m, m) and arithmetic genus $(m-1)^2$. A third pencil carried by an integral $(a, b, 1)$ surface in $(\mathbb{P}^1)^3$ is the ratio of coprime (a, b) forms on Q . Resolve its entire base cluster. With base multiplicities $n_i \geq 1$, successive curve

multiplicities $r_i \geq 0$, and total-transform exceptional classes E_i , its fibre is $aH_1 + bH_2 - \sum n_i E_i$. Therefore

$$\sum n_i^2 = 2ab, \quad T := \sum n_i r_i = (a + b - 1)m, \quad (m - 1)^2 - g \geq \sum_i \binom{r_i}{2}.$$

Cauchy–Schwarz and $\sum r_i \leq T$ prove

$$\boxed{g \leq (m - 1)^2 - \frac{T^2}{4ab} + \frac{T}{2}}. \quad (5.2)$$

Infinitely near points, repeated base points and singular graph surfaces are included. For a dependent third, $(a, b) = (1, 1)$: there are exactly two resolved simple base centres, with $r + s = m$, and cost $\binom{r}{2} + \binom{s}{2}$. Distinct pencils have distinct *complete* base clusters, including tangent data.

Theorem 5.1 (Intersection-matrix form). *Let $S \subset (\mathbb{P}^1)^3$ be an integral surface of positive type (a, b, c) and let $B \subset S$ not be contained in its singular locus. For its coordinate degrees $v = (m_1, m_2, m_3)^t$, put*

$$M = \begin{pmatrix} 0 & c & b \\ c & 0 & a \\ b & a & 0 \end{pmatrix}, \quad M^{-1} = \frac{1}{2abc} \begin{pmatrix} -a^2 & ab & ac \\ ab & -b^2 & bc \\ ac & bc & -c^2 \end{pmatrix}.$$

Then

$$g(\tilde{B}) \leq 1 + \frac{1}{2} \left[v^t M^{-1} v + (a - 2)m_1 + (b - 2)m_2 + (c - 2)m_3 \right]. \quad (5.3)$$

Proof. Let $\pi : Y \rightarrow S$ be normalization followed by its minimal resolution. For compatible canonical divisors, $D = K_Y - \pi^* K_S \leq 0$. Indeed finite duality identifies the normalized dualizing module with the conductor times a dualizing generator, so all nonexceptional coefficients of D are nonpositive. Write $D = A - N$ with effective disjoint positive and negative parts. A component of A is exceptional. For any exceptional irreducible E , minimality and adjunction give $D \cdot E = 2p_a(E) - 2 - E^2 \geq 0$. If $A \neq 0$, negative definiteness would give $0 \leq D \cdot A = A^2 - N \cdot A < 0$, a contradiction.

The coordinate fibre classes F_i on Y have intersection matrix M . Its span contains the nef big class $\sum F_i$. If Γ is the strict transform of B , Hodge index applied to $\Gamma - \sum (M^{-1}v)_i F_i$ gives $\Gamma^2 \leq v^t M^{-1} v$. The negative relative canonical divisor is supported over the singular or conductor locus, so $\Gamma \cdot K_Y \leq \Gamma \cdot \pi^* K_S$. Use adjunction and $K_S = \sum (a_i - 2)F_i$. $\square$

For equal degrees, the decisive specializations are

Surface type	genus bound	value at $m = 24$
$(2, 1, 1)$	$m^2/2 - m + 1$	265
$(2, 2, 1)$	$7m^2/16 - m/2 + 1$	241
$(2, 2, 2)$	$3m^2/8 + 1$	217

For $c = 1$, formula (5.3) is precisely the expanded weighted bound (5.2). The graph proof also covers curves in the graph surface’s singular locus. In a $(2, 2, 2)$ surface, a curve in the singular locus lies on a nonzero partial derivative of type $(2, 2, 1)$, up to permutation. Under a contrary high-genus hypothesis this reduces to the graph cases, justifying use of (5.3), including the degree-eight boundary.

5.3 The sharp independent-triple theorem

Theorem 5.2. *Let $m \geq 8$. Three degree- m pencils, every pair birational, whose eight product sections are independent satisfy*

$$g(C) \leq \lfloor A(m) \rfloor, \quad A(m) = m^2/2 - m + 1.$$

Equality occurs for every even $m \geq 8$.

Proof. Suppose $g > A(m)$. The Segre image has degree $3m$ in $\mathbb{P}^7$. If the product bundle had a ninth section, the corresponding $\mathbb{P}^8$ model lies, by Petrakiev's π_2 theorem in its valid range, on an integral surface of degree at most eight: $\pi_2(3m, 8) \leq A(m)$. Projection to $\mathbb{P}^7$ is a surface, not a curve of degree at most eight. Every Segre quadric contains it by Bézout, since $3m > 16$. Its positive type is $(1, 1, 1)$ or $(2, 1, 1)$. Independence excludes the former; (5.2) excludes the latter. Hence the product series has exactly eight sections and the image is linearly normal.

For a general hyperplane section $\Gamma \subset \mathbb{P}^6$, if $h_\Gamma(2) \leq 18$, then $h_B(2) = 8 + h_\Gamma(2) \leq 26 < 27 = h_{(\mathbb{P}^1)^3}(2)$. An extra quadric gives an integral surface component of positive type at most $(2, 2, 2)$. Independence, (5.2), (5.3) and the preceding singular-locus argument exclude all such components. Thus $h_\Gamma(2) \geq 19$. Uniform position gives

$$h_\Gamma(i + j) \geq \min(3m, h_\Gamma(i) + h_\Gamma(j) - 1), \quad h_\Gamma(1) = 7,$$

and therefore $h_\Gamma(2j) \geq \min(3m, 18j + 1)$ and $h_\Gamma(2j + 1) \geq \min(3m, 18j + 7)$. Summing the deficits bounds g by $\lfloor A(m) \rfloor$, a contradiction. For $m = 6k + r$, $0 \leq r \leq 5$, the sums are respectively

$$18k^2 - 8k + 1, \quad 18k^2 - 2k, \quad 18k^2 + 4k, \quad 18k^2 + 10k + 2, \quad 18k^2 + 16k + 5, \quad 18k^2 + 22k + 8.$$

For sharpness, on a smooth $(2, 1, 1)$ del Pezzo surface take a general curve in $|-kK|$, $m = 2k$. It has genus $2k^2 - 2k + 1$, balanced coordinate degrees, independent products and pairwise birational projections. $\square$

The refined number used here is, for $d - 1 = (N - 1 + \alpha)q + e$,

$$\pi_\alpha(d, N) = (N - 1 + \alpha) \binom{q}{2} + q(e + \alpha) + \max\{0, \lfloor (\alpha - N + 2 + e)/2 \rfloor\}.$$

Petrakiev's π_2 surface theorem is invoked in $\mathbb{P}^8$, not $\mathbb{P}^7$; its ambient-dimension hypothesis matters [4]. The same geometry gives independent-triple genus bounds 9, 14, 19 in degrees 5, 6, 7.

Theorem 5.3 (The retained genus threshold). *Every faithful A_7 -curve of genus at least 266 has gonality at least 25.*

Proof. Suppose its gonality is $m \leq 24$. Orbit fields generate; (5.1) is at most 265 for all relevant proper divisors, so every pair is birational, and $18 \leq m \leq 24$. Theorem 5.2 forces all third pencils to be dependent. Fix a pair; there are at least 33 thirds. In the simultaneous base-cluster graph, an edge has endpoint multiplicities summing to m and total defect budget $\delta = (m - 1)^2 - g$. This budget is less than twice the least edge cost. Thus no two edges are disjoint; a simple graph with that property is a star or a triangle. There are too many edges for a triangle.

If the common centre has multiplicity r , the star costs at least $\binom{r}{2} + 33 \binom{m-r}{2}$. Checking $18 \leq m \leq 24$ excludes $m \leq 23$; only $(m, r) = (24, 23)$ remains. That centre is proper, since an infinitely near centre would require a second multiplicity-at-least-23 ancestor, exceeding the budget. The system $(1, 1) - E$ sends the curve birationally to a plane curve of degree 25. A singular point would give a pencil of degree at most 23, so the plane curve is smooth of genus 276. But (3.1) gives $g \equiv 1 \pmod{3}$ for every faithful A_7 action. This contradiction finishes the proof. $\square$

The same calculation yields the following retained lower bounds. The genus-136 borderline is a smooth plane curve of degree 18, excluded by Harui's bound $|\text{Aut}| \leq 6 \cdot 18^2 < 2520$ [6].

Genus	136	169	199	211	241	≥ 266
Algebraic gonality lower bound	18	20	21	22	23	25

The seven signatures below 266 remain relevant to *gonality* 25, but no longer to the accessory theorem of Section 3.

5.4 Picard closure as a separate mechanism

One earlier mechanism is worth retaining. Suppose $g > 257$, $\text{gon}(C) = 18$, every pair of minimum pencils is birational, and $S = W_{18}^1(C)$ has at least seven elements. The pencil trick identifies the product kernel for three distinct classes A, B, D with $H^0(A + B - D)$. A kernel of dimension at most one would give a degree-54 model in $\mathbb{P}^6$ of genus at most 255. Hence $A + B - D \in S$. Distinct-triple closure makes S a subgroup coset $A + H \subset \text{Pic}^{18}(C)$.

If $2(B - A) \neq 0$, the pencils $B, 2A - B$ give a degree-36 birational quadric model from $2A$. A fifth section is impossible by Castelnuovo ($\pi(36, 4) = 187$). The containing quadric is unique, so its two rulings allow at most two elements of $H \setminus H[2]$. If that set were nonempty, a coset of $H[2]$ and the order-two kernel of doubling would force $|H| \leq 4$, a contradiction. Thus $H = H[2]$ is finite. For an A_7 -stable S , summing its classes gives invariant degree $18 \cdot 2^a$, which is incompatible with free C_5 at $(2, 7, 7)$ or free C_7 at $(3, 4, 5)$. This is a Picard-group closure argument, distinct from local ramification transport.

6 Torsion collisions and two simultaneous norms

This section records the exact older arithmetic reduction, including its corrected moving case. It is no longer needed to prove $\text{gon}(C/\tau) \geq 10$. Fix $\tau = (12)(34)$, $z = (13)(24)$, $\gamma = (567)$, $\sigma = (34)(57)$, $V_4 = \langle \tau, z \rangle$, $J = C_{A_7}(\tau)$ and $K = (S_4(1234) \times S_3(567)) \cap A_7$.

Curve	subgroup quotient	genus	Curve	subgroup quotient	genus
D	$\langle \tau \rangle$	64	F	$V_4 \times \langle \gamma \rangle$	10
E	$\langle \tau, \gamma \rangle$	22	B	J	3
Q	V_4	28	Y	$A_4(1234) \times \langle \gamma \rangle$	4
			X	K	0

There are cartesian squares after normalization: $D \xrightarrow{q} E$, $Q \rightarrow F$ are etale degree three, with vertical degree-two maps induced by z ; and $F \xrightarrow{\pi} Y$, $B \xrightarrow{\varphi} X$ have degree three, with vertical double covers $p : F \rightarrow B$, $h : Y \rightarrow X$. Here $F \rightarrow X$ is the S_3 closure of the trigonal map φ , π is etale, and $Q \rightarrow Y$ is an etale C_3^2 cover. These definitions distinguish the genus-22 E here from the genus-one quotient in Section 8.

The fixed-point counts give $\text{gon}(E), \text{gon}(F) \geq 4$: a genus-at-least-five trigonal curve has a unique trigonal pencil and an involution fixes at most six points; a nonhyperelliptic involution on a hyperelliptic curve fixes at most four. The relevant involutions fix eighteen and ten points. Consequently effective degree-three classes on E, F have unique representatives.

Let $\eta \in \text{Jac}(E)[3]$ specify q , and let $\xi \in \text{Jac}(F)[3]$ satisfy $\beta^* \xi = \eta$, where $\beta : E \rightarrow F$. The actual double-cover root $\lambda \in \text{Pic}^3(F)$ satisfies $2\lambda = B_F$, $\beta^* \lambda = R_z^E$. For any etale cyclic cover of degree ℓ , a nonzero character collision $[U - V] = \eta^j$ pulls back to distinct equivalent degree- $\ell \deg U$ divisors;

after cancellation it gives a pencil of at most that degree. Thus transport proves

$$W_3(E) \cap (W_3(E) + \eta) = \emptyset. \quad (6.1)$$

Conversely, in the hypothetical minimal nine-pencil case, there are at most two such pencils: birational pairs and the three-pencil bound, or a common index-three genus-four core with at most two trigonal pencils, prove this. The deck C_3 preserves each pencil and acts nontrivially on its target. Its two fixed fibres give the exact dictionary with ordered pairs $(U, V) \in \text{Sym}^3(E)^2$, $[U - V] = \eta$.

For $M = \mathcal{O}_E(U)$, saturation and the double-cover norm imply

$$\text{Nm}_\beta M \in \{\lambda, \lambda + \xi, \lambda + 2\xi\}. \quad (6.2)$$

Each effective root has one representative and at most eight degree-three lifts, so this is a finite test of at most 24 candidates for $h^0(E, \mathcal{O}_E(U) - \eta) > 0$. Effectivity of a root alone is not the collision test.

The V_4 cover $E \rightarrow B$ has disjoint labelled branch divisors $\deg \Delta_z = 3$, $\deg \Delta_\sigma = 9$, $\deg \Delta_{z\sigma} = 1$. Write $A = \varphi^* \mathcal{O}_X(1)$ and $K_B \sim A + P_0$. The canonical model of B is a plane quartic. Its ten simply branched trigonal fibres are $2R_i + T_i$, giving

$$R_\varphi \sim K_B + 2A, \quad \Delta_\sigma + \Delta_{z\sigma} \sim 6A - 2K_B, \quad \text{root}(F/B) = 3A - K_B.$$

Two distinguished fibres have the form

$$\varphi^{-1}(x_0) = P_1 + P_2 + P_\tau, \quad \Delta_z = P_1 + P_2 + P_3, \quad \varphi^{-1}(b_0) = 2P_3 + P_t, \quad \Delta_{z\sigma} = P_t.$$

The quartic geometry gives $h^0(\Delta_z) = 1$ if $P_3 \neq P_0$, and two otherwise. Let $y_\pm$ lie over x_0 , let y_* be ramified over b_0 , and put $\Theta = y_+ + y_- + y_*$. Then every root in (6.2) obeys *both* norm identities

$$\text{Nm}_p(\lambda + k\xi) = \mathcal{O}_B(\Delta_z), \quad \text{Nm}_\pi(\lambda + k\xi) = \mathcal{O}_Y(\Theta), \quad k = 0, 1, 2. \quad (6.3)$$

These are identities of actual roots, not just their squares. If M_2 is the actual root of $E/\sigma \rightarrow B$ and s_t lies over P_t , then $\lambda = p^*M_2 - s_t$, $2M_2 = \Delta_z + P_t$. This proves the first norm; eigensheaf multiplication and the coprimality of two and three prove the second. Also $|\Theta| = \{y_* + h^{-1}(u) : u \in X\}$.

In the rigid case $P_3 \neq P_0$, simultaneous norms leave four lifted triangles; λ is ineffective, and a collision must pair the two nonzero twists. It reduces to $[U - \sigma U] = \eta$, two tests up to sign. Equation (6.1) excludes them. In the moving case $P_3 = P_0$, with $p^{-1}(P_3) = \{s_+, s_-\}$, there are *three* compatible divisors,

$$2s_+ + s_t, \quad s_+ + s_- + s_t, \quad 2s_- + s_t.$$

The middle can represent λ ; the others can represent its two nonzero twists. Discarding the latter pair was the earlier error. Untwisted effectivity still satisfies λ effective $\iff M_2 \sim P_3 + P_t$.

The signed form of a compatible collision is $\sum_{i=1}^3 \epsilon_i [p_i - \sigma p_i] = \pm \eta$, in a Prym of dimension 15. If it held, $\text{div}(u) = 3U - 3\sigma U$ and $y^3 = u$ reconstruct the specified cyclic cover; the target actions can be written $\gamma : y \mapsto \zeta_3 y$, $z : y \mapsto -y$, $\sigma : y \mapsto 1/y$. For an H -preserved nine-pencil the additional quartic relation is $3P_\tau \sim P_3 + 2P_0$. It is a marked flex condition, not an automatic consequence of norm effectivity. The full A_7 structure is indispensable: a smooth $(9, 9)$ curve with the same $C_2 \times S_3$ fixed-point data exists, so those local data alone cannot exclude nine-gonality.

7 Spectral and conformal frameworks

7.1 A metric-independent gonality inequality

For a conformal metric g and a degree- d holomorphic map $x : C \rightarrow S^2$, Hersch balancing makes its three coordinates mean zero. Their total Dirichlet energy is $8\pi d$, while $\sum x_i^2 = 1$. Rayleigh

therefore gives

$$\lambda_1(g) \text{Area}(g) \leq 8\pi d. \quad (7.1)$$

For hyperbolic C of genus 136, this yields $\text{gon}(C) \geq 67.5\lambda_1(C)$ and $\text{gon}(C/\tau) \geq 33.75\lambda_1(C)$. More generally define

$$\Lambda^G(C) = \sup_{h>0, G\text{-invariant}} \lambda_1(hg) \int_C h dA_g \leq 8\pi \text{gon}(C).$$

Only the mass form changes under $g \mapsto hg$; the Dirichlet form does not.

7.2 Representation coverage and certified finite elements

The exact surface is tiled by 5040 hyperbolic triangles of angles $\pi/2, \pi/4, \pi/7$. In a common Klein chart, write $x = X_A + J(s, t)^t$ on $R = \{s, t \geq 0 : s + t \leq 1\}$. The coefficients are

$$A_R = |\det J| J^{-1} \frac{I - xx^t}{\sqrt{1 - |x|^2}} J^{-t}, \quad w_R = \frac{|\det J|}{(1 - |x|^2)^{3/2}}.$$

The gluings preserve the reference coordinates; the proof does not replace the surface by flat triangles. For a subgroup K with sign character ϵ , use functions $f(kx) = \epsilon(k)f(x)$. Every eigenvalue is seen if its irreducible representation contains one of the chosen restricted characters.

Problem	subgroup	index	irreducibles detected
Q_0	A_7	1	1 (remove constants)
Q_1	$C_3^2 : C_4$	70	6, 14_a , 14_b , 15, 21
Q_2	$C_{A_7}(\tau)$	105	10, $\overline{10}$, 15, 35 (35 twice)

Character averaging $|K|^{-1} \sum_{k \in K} \epsilon(k) \chi(k)$ proves coverage. Thus

$$\lambda_1(C) \geq \min\{\mu_2(Q_0), \mu_1(Q_1), \mu_1(Q_2)\}.$$

The actual Q_2 group is $C_3 \rtimes D_8$, of order 24, with element counts $1 : 1, 2 : 9, 3 : 2, 4 : 6, 6 : 6$ and center C_2 . It is not S_4 .

On each reference cell, rigorous coefficient bounds choose constant $A_e \preceq A_R$ and $w_e \geq w_R$. Let a, b denote comparison energy and mass, and let Π be the Crouzeix–Raviart edge-mean interpolant. Integration by parts gives

$$a_h(u - \Pi u, v_h) = 0, \quad a(u) = a_h(\Pi u) + a_h(u - \Pi u), \quad \|u - \Pi u\|_b \leq C_h \sqrt{a_h(u - \Pi u)}.$$

On the uniform mesh, $C_h^2 \leq \max_e \frac{w_e}{\lambda_{\min}(A_e)n^2} (\frac{1}{8} + \frac{2}{j_{1,1}^2})$. If $K_h - \sigma M_h \succ 0$, triangle inequality and Cauchy–Schwarz prove

$$\mu_1 \geq \frac{\sigma}{1 + C_h^2 \sigma}. \quad (7.2)$$

For Q_0 , subtract the comparison mean from Πu and use weighted variance $\min_c \|u - c\|_b^2$. This handles the different true and comparison mean-zero spaces.

An a posteriori matrix certificate avoids assuming that a numerical factorization is accurate. If $\|B - \hat{B}\|_2 \leq \eta$, and exported L, P satisfy

$$\|LL^t - P(\hat{B} - cI)P^t\|_2 \leq \delta, \quad c > \eta + \delta,$$

then $B \succeq (c - \eta - \delta)I \succ 0$. The residual can be enclosed through row/column norms and rational binary64 error bounds, including sparse products, reductions, coalescing, permutations and assembly.

For unit roundoff u , use $\gamma_k = ku/(1-ku)$; one useful enclosure is $\delta \leq \rho/(1-u) + \gamma_{k+1}\|L\|_1\|L\|_\infty$. The interpolation input $j_{1,1} > 19/5$ follows directly from the alternating Bessel series: for $y \leq 361/100$,

$$\frac{J_1(x)}{x/2} \geq 1 - y/2 + y^2/12 - y^3/144 + y^4/2880 - y^5/86400 > 0.$$

Numerical status. The earlier audited certificate reports $\lambda_1 > 0.34089$, hence $\text{gon}(C) \geq 24$ and $\text{gon}(C/\tau) \geq 12$. The recovered PR's independent rerun records eleven of twelve cases complete; the last large Q_1 case and the full export bundle are not certified by this reference. The abstract proof, coefficient inequalities and factor-residual criterion are retained; no rounded table is substituted for the missing standalone numerical data. The Lean files verify conditional analytic and arithmetic implications, not the external numerical data automatically. None of this numerical status affects Section 3.

7.3 Strict improvement and the actual optimizer

If the first real eigenspace is an absolutely irreducible representation of dimension q , use an orthonormal frame φ_i , area A , and $\bar{F} = (A/q) \sum \varphi_i^2$. For invariant u ,

$$\left. \frac{d}{dt} \right|_0 [\lambda_1((1+tu)g) \text{Area}((1+tu)g)] = \lambda_1 \int_C u(1-\bar{F}) dA. \quad (7.3)$$

This is a local first-variation statement; the asserted 14_a first eigenspace in the computational models is separate from its abstract derivation.

For a hyperbolic triangle action, no invariant positive-semidefinite frame on the first eigenspace has constant squared length one. Such a frame would give a harmonic map to a sphere with constant positive energy. Its Hopf differential is invariant and holomorphic, hence zero, because a triangle orbifold has no such quadratic differential. It would be a minimal immersion of constant negative curvature into a sphere, excluded by Bryant [9]. Compactness and separation of the full normalized cone of invariant positive-semidefinite frames then produce a conformal variation raising every lowest eigenvalue branch. Thus the invariant conformal gain is strict even when the eigenspace is reducible. It gives no quantitative percentage.

A useful quantitative diagnostic, when the first eigenspace is irreducible, is the following. If a degree- d map exists, put $\zeta_d = (8\pi d/A - \lambda_1)/(\lambda_{\text{next}} - \lambda_1)$. Projection of its balanced coordinate vector onto the first eigenspace and group averaging give

$$\frac{1}{A} \int_C |\bar{F} - 1| \leq 2\zeta_d + 2\sqrt{\zeta_d}.$$

Indeed the discarded squared norm is at most $A\zeta_d$; Cauchy–Schwarz controls the cross term. Harmonicity also gives the exact identity $P_1((|\nabla x|^2 - \lambda_1)x) = 0$.

There is now a precise existence statement for S4. Vinokurov's equivariant optimization theorem, Corollary 1.4, gives attainment for the first eigenvalue when every group orbit has size greater than one [10]. Attainment is in the natural class allowing isolated conical singularities; a smooth positive maximizer does not follow. On the $(2, 4, 7)$ curves the orbit sizes are 360, 630, 1260, 2520, so it applies. An optimizer comes from an equivariant harmonic map into *some* sphere; the triangle Hopf argument makes it branched conformal. The target representation need not stay 14_a , and no argument confines its branch or cone points to the three special orbits. For a fixed target, the elliptic deformation problem has a finite-dimensional local kernel. That is not rigidity, a global finite list, or an exact formula for the optimum. The necessary Jacobi-field and spectral-crossing analysis is open.

The remaining sufficient spectral thresholds for gonality 25 are

Signature	genus	inner marked classes	strict λ_1 threshold
(2, 4, 7)	136	4	16/45
(3, 3, 5)	169	2	2/7
(2, 5, 7)	199	4	8/33
(3, 3, 6)	211	2	8/35
(3, 4, 4)	211	8	8/35
(2, 6, 7)	241	4	1/5
(3, 3, 7)	241	4	1/5

These are marked-cover counts, not counts of distinct underlying curves. The rule is $\lambda_1 > 48/(g-1)$. For the numerical baseline 0.34627, the (2, 4, 7) gain needed is about 2.7%; from 0.34089 it is about 4.30%.

8 Exact arithmetic models and Abel–Jacobi carriers

8.1 The septic Belyi family

Let $s = 1 \pm 4\sqrt{21}/21$, $t = 2s/3 - 10/21$, so $21s^2 - 42s + 5 = 0$, and set

$$q_s(z) = \frac{z^7}{7} - \frac{(s+1)z^6}{6} + \frac{(s+t)z^5}{5} - \frac{tz^4}{4}, \quad r_s = -\frac{128(51s-65)}{1750329}, \quad \beta_s(z) = 1 - q_s(z)/r_s.$$

Then $q'_s = z^3(z-1)(z^2-sz+t)$, $q_s(0) = q_s(1) = 0$, and the two quadratic critical points have value r_s . The passport is $[2^2 1^3, (4, 2, 1), 7]$, and

$$\text{disc}_z(\beta_s(z) - T) = -7(-1/r_s)^6 T^2 (T-1)^4.$$

The equal-critical-value condition factors as $(7s^2 - 7s + 4)(21s^2 - 56s + 40)(21s^2 - 42s + 5)$. The middle factor degenerates; the first is the $\sqrt{-7}$ Klein case. For the last, specialization modulo 17 at $s = 3$, $\beta_s(z) = 2$ has factor degrees two and five. Its monodromy cannot lie in the normalizer of the degree-seven $\text{PSL}_2(7)$ group. The maximal-subgroup list then forces A_7 geometrically. The Belyi maps are defined over $\mathbb{Q}(\sqrt{21})$; the full marked regular actions can be defined over $\mathbb{Q}(\sqrt{21}, \sqrt{-7})$. Minimal fields for the unmarked curves are not determined by this statement.

8.2 Two resolvents

Make $(\beta_s(X) - \beta_s(z))/(X - z)$ monic with roots $r_1, \dots, r_6$. For a perfect matching M of these six labels put $\sigma_M = \sum_{\{i,j\} \in M} r_i r_j$. The six pentads are the partitions of the fifteen edges into five perfect matchings. Define

$$u_\Pi = \sum_{M \in \Pi} \sigma_M^3, \quad R_{\mathcal{E}}(U, z) = \prod_{\Pi} (U - u_\Pi).$$

Its normalized curve is the genus-one quotient $\mathcal{E} = C/A_5$ for the transitive six-letter A_5 . The cube is essential: the analogous first and second power sums are constant. For the ten unordered partitions $A \mid A^c$ into triples put

$$v_A = \prod_{i \in A} r_i + \prod_{i \in A^c} r_i, \quad R_{\mathcal{S}}(V, z) = \prod_{A \mid A^c} (V - v_A).$$

Its normalization is the genus-two quotient $C/(C_3^2 : C_4)$. The invariants generically separate the required cosets: modulo 457, at $s = 118, z = 29$, the six pentad values are 128, 123, 375, 88, 438, 302, and the ten partition values are 130, 40, 334, 449, 301, 390, 185, 428, 395, 391, all distinct. An integral version uses $\lambda = 21s$, $\lambda^2 - 42\lambda + 105 = 0$, $Z = 18z$, and

$$P_\lambda(X) = X^7 - (\lambda + 21)X^6 + 36(\lambda - 6)X^5 - 324(\lambda - 15)X^4.$$

The monic sextic $(P_\lambda(X) - P_\lambda(Z))/(X - Z)$ gives the same resolvent construction. These are exact function-field specifications, not expanded Weierstrass or hyperelliptic equations with marked point coordinates.

8.3 The two rational isotypic obstructions

Chevalley–Weil gives

$$\text{Jac}(C) \sim A^{10} \times E_1^{15} \times E_2^{21} \times S^{35},$$

where $\dim A = 3$, E_i are elliptic, $\dim S = 2$, and A has $\mathbb{Q}(\sqrt{-7})$ action. The involution-fixed-divisor permutation module is

$$\mathbb{Q}[A_7/C_G(\tau)] \simeq 1 + 6 + 14_a + 2 \cdot 14_b + 21 + 35.$$

Only the unique source copies of 21 and 35 occur in the Jacobian representation. After tensoring the point group with $\mathbb{Q}$, Abel–Jacobi on either simple source copy is zero or injective. Both the Klein difference and the lift of $\Delta_\sigma - 3\Delta_z$ have nonzero components in both copies. Consequently the two classes are torsion together.

The quotient $\mathcal{E}$ detects the 21-part; all 42 conjugates of its A_5 subgroup detect the Klein difference. The genus-two quotient detects the 35-part; 58 of its 70 conjugates detect the class, so a detecting one must be selected. Proving either pushed-forward class non-torsion settles both torsion questions. Until an origin is supplied on $\mathcal{E}$, the correct elliptic group is $\text{Jac}(\mathcal{E})$, not an assumed pointed genus-one curve.

No CM determination, conductor, rank, or torsion order has been computed. The arithmeticity of the triangle group does not force CM of this elliptic quotient. A rank calculation also does not alone decide whether a specified point is torsion. Once a model and the point are explicit, exact multiplication with torsion bounds from good reductions, or a positive certified canonical height, can decide this without first knowing the full rank. A modular L -function rank computation requires an identified elliptic curve and the relevant arithmetic hypotheses.

The triangle group $(2, 4, 7)$ is arithmetic (Takeuchi), correcting the old contrary sentence in the literature notes. This compact triangle curve does not fall under the cuspidal Manin–Drinfeld theorem. No general torsion theorem for its elliptic-point differences is being invoked.

9 Structural answers and the remaining ladder

S1: the program is not simply $\gamma(A_7)$. Ordinary gonality has the recovered upper bound 56: for a Sylow 2-subgroup P of order eight, $g(C/P) = 12$, and the general genus-12 bound supplies a degree-seven pencil. Its pullback has degree 56 (and degree 28 on C/τ). Yet the accessory threshold is 60. The structural replacement is (1.1), with linearized moving degree and subgroup induction. Even for A_5 , $\gamma(A_5) = 1$ while $a(A_5) = 2$. There is no general converse from a degree- d pencil to a degree- d accessory. For a connected full- A_7 compression to a fixed $(2, 4, 7)$ target, Theorem 1.2 and (2.1) even require $90 \mid d$. Whether an accidental equivalence holds in a restricted range below a chosen wall cannot be inferred without computing the actual gonality.

S2: the lattice is standard; transport supplies the class. Formula (2.2) confirms both stated numerical contradictions. It is weaker in general than the exact linearized lattice and can be weaker than cyclic restriction. Transport includes the nontrivial construction of an invariant class by local saturation and normalizer generation. One can define the degree subgroup of invariant elements in a specified integral Picard module generated by the pencil orbit and ramification classes; calculating its relations is the substance, not an automatic extension beyond saturation. Equation (4.2) records the missing residual condition. The degree-ten dimension comparison, seven parameters against a 45-dimensional wrong-isotype Jacobian factor, proves no emptiness for the specified fibre.

S3: global orbit geometry remains an independent problem. Sections 5–6 are a multigraded Castelnuovo theory adapted to triples and simultaneous base clusters. They use the whole orbit to force a star, but no theorem here bounds every transitive A_7 orbit of 35 pencils below genus 136. The extremal degree-four del Pezzo examples prove that a triple-only improvement would be false. Additional orbit syzygies, compatibility of surface envelopes, or equivariant Hilbert functions would be needed. The accessory theorem bypasses this remaining gonality question by imposing a linearized complete series.

S4: existence is known here; exact rigidity is not. Section 7 gives possibly conical attainment with an equivariant harmonic map, using the no-fixed-point corollary. Neither the target S^{13} , the representation 14_a , branch support, uniqueness, nor an exact optimum is established. A finite-dimensional local deformation problem is not a finite classification.

S5: the arithmetic problem is now well posed but unfinished. Section 8 supplies exact quotient function fields and the correct Jacobian carriers. The next concrete step is normalization into explicit low-genus models with the images of the ramification divisors. CM, conductor, rank and torsion need those models; none is implied by the quaternionic origin alone.

S6: different intrinsic quantities control different questions. For a nonabelian simple group with minimum faithful genus $g_{\min}$, the orbit-field argument gives

$$\gamma(G) \geq \min\{|G|, \lceil 1 + \sqrt{g_{\min}} \rceil\}.$$

On the other hand $\gamma(G) \leq \lfloor (g_{\min} + 3)/2 \rfloor$ by the general gonality bound applied to a minimum-genus curve. These leave a wide range; square-root growth is a lower-bound scale, not a proved asymptotic law. The Valentiner smooth plane sextic gives $\gamma(A_6) = 5$: its genus is ten, the minimum possible by Riemann–Hurwitz with inertia orders 2, 3, 4, 5, and a smooth plane sextic has gonality five. It should not be confused with the singular genus-six Wiman sextic with S_5 symmetry [12].

For $\mu(G) < |G|$, Proposition 2.2 instead forces $g_{\min} \leq \pi(\mu(G), q(G) - 1)$, roughly $\mu(G) \gtrsim \sqrt{2(q(G) - 2)g_{\min}}$, before the ramification lattice is imposed. Thus the honest linear representation degree enters this stronger obstruction. For A_n , Hurwitz gives $g_{\min} \geq 1 + |A_n|/84$ and the usual minimum linear degree is $n - 1$ for $n \geq 7$. These are intrinsic lower-bound mechanisms, not an identification with ordinary essential dimension. Subgroup induction then controls $a(G)$, as the index-15 route does for A_7 .

The next mathematical targets. The bounded-accessory target is complete at 60 in the stated convention. The independent open problems are: the exact $\gamma(A_7)$ and the seven gonality-25 cases; the fixed full- A_7 value $\mu(A_7)$ beyond its lower bound 60; explicit arithmetic models and the two torsion classes; and a structure that survives unrestricted resolvent towers. The correspondence argument uses a fixed point of an affine linear base and $\text{Pic}(V \times C) = \text{Pic}(C)$. Intermediate accessory varieties do not automatically retain these properties. Consequently the proof has no presently established iteration law for arbitrary one- or two-variable towers, and proves neither $\text{RD}(A_7) > 1$ nor Hilbert’s thirteenth conjecture.

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